



National University of Technology (NUTECH)

Computer Science Department

Probability and Statistics (MT220)

Probability and Statistics (MT220)

Contact Hours

Theory = 3/week

Total = 48**Credit Hours**

Theory = 3

Total = 3

Pre-requisite: None in probability and statistics, but familiarity with elementary linear algebra (vectors, matrices) and calculus (derivatives, integrals) is assumed.

COURSE DESCRIPTION:

This course gives an introduction to probability and statistics, introducing students to quantitative uncertainty analysis and risk assessment with emphasis on engineering applications. The course focuses on probability theory and its applications, with a module covering basic topics in statistics (Basic Statistics, samples, sampling distributions, parameter estimation, hypothesis testing and regression analysis). The probability modules cover events and their probability, the total probability and Bayes' theorems, discrete and continuous random variables and distributions, cumulative distribution function, conditional distributions, moment generating function. Throughout the subject, concepts are illustrated with examples from engineering and everyday life.

COURSE OBJECTIVES:

1. To provide students a deeper understanding of the basic concepts of probability, statistics and the random variables and familiarize students with the probability models related to engineering.
2. To provide students sufficient knowledge and skills in solving relevant problems in engineering and architecture.
3. To strengthen and broaden students' knowledge in probability concepts and random variables which they may encounter in solving different types of engineering problems.

COURSE LEARNING OUTCOMES:

Upon successful completion of the course, student should be able to:

S#	CLO	Learning Domain	Taxonomy Level	PLO
1	Explain the basic concepts of statistics and probability	Cognitive	2	1
2	Apply the probability models to solve various engineering and real life problems.	Cognitive	3	1
3	Analyse the data using statistical inference for decision making	Cognitive	4	1

	National University of Technology (NUTECH) Computer Science Department
	<u>Probability and Statistics</u> <u>(MT220)</u>

RELEVANT PROGRAM LEARNING OUTCOMES (PLOs):

The course is designed so that students will achieve the following PLOs:

1	Academic Education	✓	7	Communication	□
2	Knowledge for Solving Computing Problems	□	8	Computing Professionalism and Society	□
3	Problem Analysis	□	9	Ethics	□
4	Design/Development of Solution	□	10	Life-long Learning	□
5	Modern Tool Usage	□			
6	Individual and Team Work	□			

INSTRUCTOR:

Name: Dr. Faryal Younis
 Office: Mechanical Faculty Office, Lab Block, 5th floor
 Email: faryal@nutech.edu.pk
 Cell: 03125597908
 Class Hours: Check weekly training programs for class timings.

TEXT AND MATERIAL:

Text books

RE Walpole, RH Myers, SL Myers, K Ye, *Probability & Statistics for Engineers & Scientists*, 9th Edition.

Reference Book

1. Richard L. Scheaffer and James T. McClave: *Probability and Statistics for Engineers*, 5th Edition
2. MH Degroot and MJ Schervish, *Probability and Statistics*, 4th Edition.

ASSESSMENT SYSTEM:

Relative grading system will be followed to award grades. %age weight of different exams is as under:

Type of Exam	No	% weightage
Assignments	4	5
Quizzes	4	10
PBL	1	10
Mid Semester Exams	1	25
End Semester Exams	1	50



National University of Technology (NUTECH)

Computer Science Department

Probability and Statistics (MT220)

CONDUCT IN THE CLASS:

- Students are not allowed to chat with each other.
- Students are not allowed to do work of any other subject during the class.
- Your cell phones should be on silent/vibration mode.
- Everyone should be seated in the class 1-2 minutes ahead of the start time of the class.
- You are not allowed to leave the class without permission.
- You are not allowed to enter class without permission after class has started.

WEEKLY PLAN:

Week	Topic Covered	Reading Assignment	CL O No.	PLO No.	Assessment Methodology
1	Introduction to statistics and data analysis (Basic Statistics, Sampling procedures, Measures of central tendency)	Chap 1: Sec 1.1-1.3	1	1	Quiz, PBL, Assignment, MSE, ESE
2	Measures of dispersion, Statistical modelling, Graphical presentation	Chap 1: Sec 1.4-1.7			
3	Probability, Additive rule, Independence	Chap 2: Sec 2.1-2.5	1, 2	1	
4	Conditional prob., Total prob. and Bayes' theorem	Chap 2: Sec 2.6, 2.7	1,2	1	
5	Random Variables and Probability distributions (PMF, PDF, CDF)	Chap 3: Sec 3.1-3.4			
6	Mathematical expectation, Mean and Variance of random variable and linear combinations, Chebyshev's theorem	Chap 4: Sec 4.1-4.4	1,2	1	
7	Joint and conditional distribution, Covariance, Coefficient of correlation	Chap 4: Sec 4.1-4.4			
8	Midterm Exam				
9	Discrete distributions (Binomial, Hypergeometric, Geometric, Negative Binomial, Poisson)	Chap 5: Sec 5.1-5.3 5.4-5.5	1,2	1	
10	Continuous distributions (Uniform, Exponential, Normal)	Chap 6: Sec 6.1,6.2, 6.4, 6.6			
11	Normal Distribution	Chap 6: Sec 6.2-6.4			



National University of Technology (NUTECH)

Computer Science Department

Probability and Statistics (MT220)

12	Statistics, Sampling and Sampling distributions, Central Limit Theorem, Introduction to inferential Statistics, Estimation of parameters	Chap 8: Sec 8.1-8.4 Chap 9: Sec 9.1-9.4, 9.10, 9.12	3	1	
13	Hypothesis testing	Chap 10: Sec 10.1-10.5			
14	Hypothesis testing	Chap 10: 10.8-10.10			
15	Simple linear regression, Curve fitting, Coefficient of correlation, Coefficient of determination	Chap 11: 11.1-11.4, Notes			
16	Problem Solving Session				
	Final Exam				

Written By: Dr. Faryal Younis

(Instructor)

Date:

Received By: Dr. Muhammad Rashid

(CS HOD)

Date:

Approve By: Muhammad Rashid

(CS HOD)

Date: